

Unsupervised Deep Clustering of Multivariate Particle Trajectories in PIC Simulations

M2 Internship Report Defense

Linshuo CHEN

Sorbonne Université
M2 – Parcours Statistique, apprentissage et algorithmes

Supervisor: Andrea CIARDI

Talk roadmap

1. Scientific motivation and data
2. Multichannel spectrogram workflow
3. Deep clustering and experiment design
4. Main results and diagnostics
5. Computational cost, limitations, and outlook

Central question

Can learned time–frequency representations organize multivariate PIC particle trajectories without predefined labels?

Why cluster particle trajectories?

- PIC simulations generate large collections of particle histories.
- Each trajectory carries multiple, temporally varying physical quantities.
- Manual inspection and a few 2D projections do not scale to thousands of particles.
- Unsupervised grouping can help organize records for exploration and follow-up analysis.

Research objective

Extend a spectrogram-based clustering workflow to six-channel particle data, then compare its latent-space organization with linear and time-series alternatives.

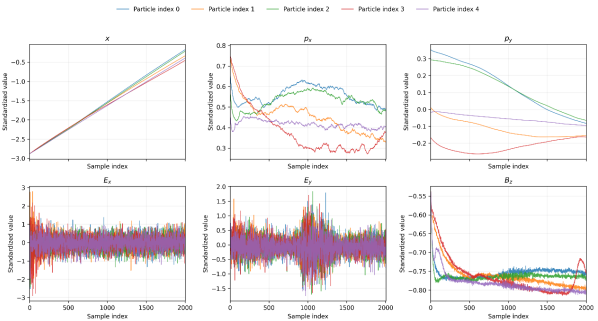
No predefined particle-class labels are available.

Dataset and six input channels

Dataset at a glance

- 12,800 tracked particles
- 6,010 recorded time points
- HDF5 trajectory records
- 2,004 samples per particle after fixed-stride subsampling

Particle IDs link predictions back to original records; they are not model inputs or labels.



Standardized examples; each color denotes one particle across channels.

Channels: x, p_x, p_y, E_x, E_y, B_z. The available p_z channel is not used.

Preprocessing: preserve correspondence, balance scale

1. Select every third recorded sample: 6,010 $\rightarrow$ 2,004 time points.
2. Apply identical time indices to all particles and all six channels.
3. Standardize each channel using statistics across the available dataset.
4. Construct either standardized time-series inputs or six-channel spectrogram inputs.

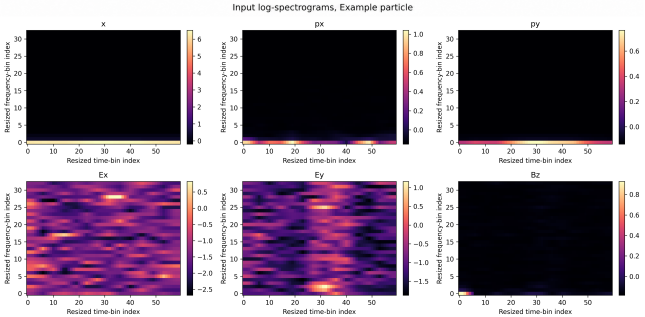
Evaluation scope

Because preprocessing statistics use the full dataset, reported scores are internal clustering assessments, not independent test-set generalization results.

From time series to multichannel spectrograms

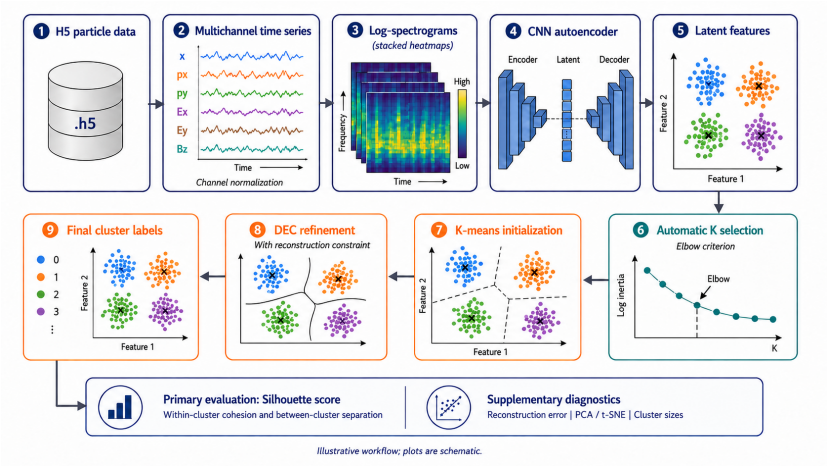
- Short-time Fourier magnitude for each standardized channel
- Window: 256 samples; hop: 128 samples
- Log compression: $S_{\log} = \log(1 + S)$
- Native grid: 129×14; resized to 33×60
- Standardize each channel; stack into $6 \times 33 \times 60$

Resizing creates a fixed input shape; it does not add independent signal information.



Example six-channel log-spectrogram input.

End-to-end workflow



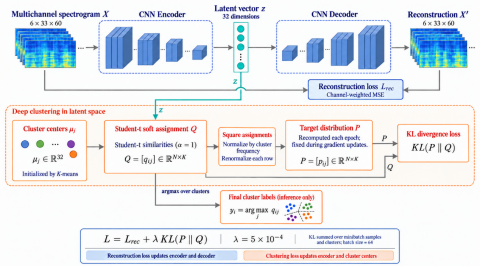
The spectrogram CNN + DEC pipeline is the principal method; PCA + K-means and 1D CNN + DEC serve as references.

Representation learning with a convolutional autoencoder

- Four 2D convolutional encoder layers: 8, 16, 32, and 64 channels
- Compress each $6 \times 33 \times 60$ input to a 32-dimensional code z_i
- Decoder reconstructs the input spectrogram
- Pretraining minimizes channel-weighted reconstruction error

Why pretrain?

Learn a compact representation before introducing cluster assignments.



Encoder, decoder, and reconstruction-constrained clustering.

DEC refinement: jointly shape features and clusters

Given latent point z_i and cluster center μ_j, DEC computes a soft assignment q_{ij}, then sharpens assignments into a target distribution p_{ij}.

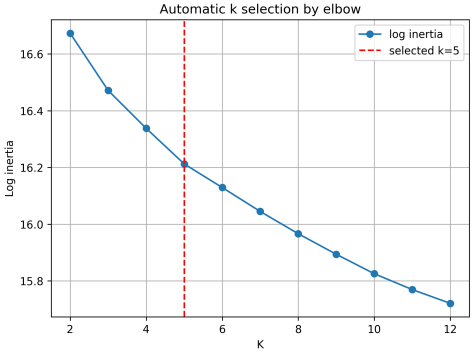
$$q_{ij} \propto \left(1 + \frac{\|z_i - \mu_j\|^2}{\alpha}\right)^{-(\alpha+1)/2}, \qquad p_{ij} \propto \frac{q_{ij}^2}{\sum_i q_{ij}}, \quad \alpha = 1.$$

Joint objective

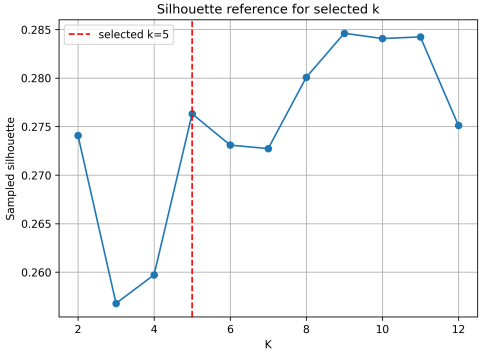
$$L = L_{\text{recon}} + \lambda L_{\text{cluster}}, \qquad L_{\text{cluster}} = \mathrm{KL}(P\|Q), \qquad \lambda = 5 \times 10^{-4}.$$

K-means initializes five centers. The reconstruction term helps retain information about the input while the latent assignments are refined.

Selecting the number of clusters



(a) Log-inertia elbow criterion.



(b) Silhouette values shown for reference.

Decision

The normalized elbow criterion selects $K = 5$ from candidates $K = 2, \ldots, 12$. The silhouette maximum is not used to choose K.

Experimental comparison

Method	Input	Representation and clustering
PCA + K-means	Six-channel spectrogram	Flatten, PCA to 32D, K-means
1D CNN + DEC	Standardized time series	1D autoencoder, 32D, DEC
Spectrogram CNN + DEC	Six-channel spectrogram	2D autoencoder, 32D, DEC

- Same 12,800-particle dataset; five clusters for all methods.
- Silhouette computed on the same 10,000 particles with Euclidean distance.
- Each method is scored in its own 32-dimensional representation space.

Main result: strongest internal silhouette for spectrogram CNN + DEC

Method	Silhouette
PCA + K-means	0.29957
1D CNN + DEC	0.79637
Spectrogram CNN + DEC	**0.87985**

Observed difference

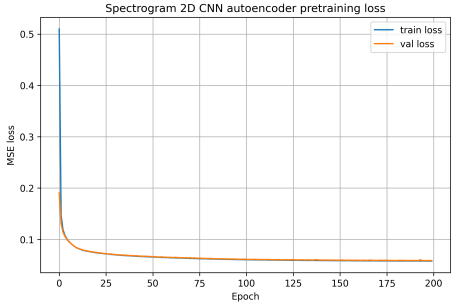
Spectrogram CNN + DEC exceeds the 1D CNN + DEC score by 0.08348 in this experiment.

0.87985

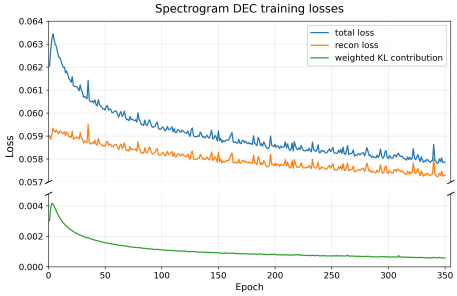
best score among tested configurations

Internal metric in each method's own latent space; not classification accuracy or a direct comparison in original trajectory space.

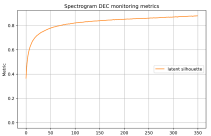
Training diagnostics: optimization and monitoring



Autoencoder pretraining: training and validation reconstruction loss.

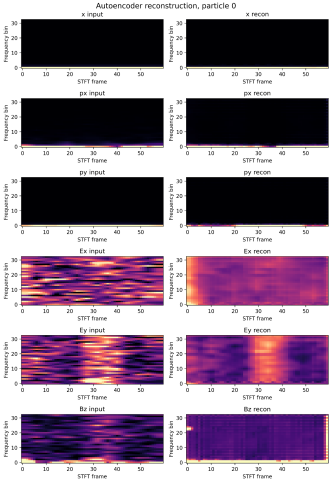


DEC refinement: reconstruction, clustering, and total losses.



Silhouette is monitored on 5,000 particles during refinement; final evaluation uses 10,000. Training was run for 350 DEC epochs.

Reconstruction: a representation diagnostic

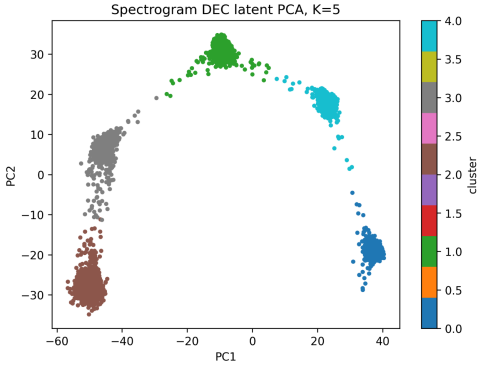


What this checks

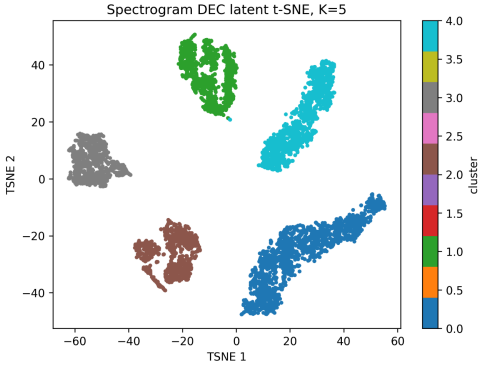
Can the pretrained autoencoder reproduce the broad channel-wise time–frequency patterns?

- Rows correspond to $x, p_x, p_y, E_x, E_y, B_z$.
- Reconstruction loss measures input fidelity.
- It is a supplementary diagnostic, not a clustering-quality score.

Final latent-space visualization



PCA projection.



t-SNE projection.

Colors show the five final labels. Clustering and silhouette are computed in 32D, not in these 2D projections.

Five groups: distribution across all particles

Cluster	Particles	Share
C0	3,884	30.34%
C1	2,314	18.08%
C2	2,068	16.16%
C3	1,818	14.20%
C4	2,716	21.22%
Total	12,800	100.00%

Interpretation

All clusters are nonempty and contain substantial numbers of particles.

Cluster indices are identifiers assigned by the algorithm, not physical classes. Size alone does not validate a partition.

Computational cost and scale-up scenario

Measured: 12,800 particles

- Spectrogram CNN + DEC: **524.62 s**
- 1D CNN + DEC: **405.19 s**
- NVIDIA RTX PRO 6000 Blackwell Server Edition

Includes preprocessing, pretraining, 350 refinement epochs, evaluation, and figure generation.

Linear-scaling estimate

- 1 million particles: about **11.4 hours**
- 10 million particles: about **4.7 days**

Projection assumes unchanged settings and per-particle dimensions. It is a rough estimate, not a benchmark at scale.

Large-scale use would require careful data loading and memory management.

Limitations and interpretation

- **Internal metric:** silhouette measures cohesion and separation in a chosen representation; it does not show physical ground-truth classes.
- **Different spaces:** each method is evaluated in its own latent space, so scores are not a common-space comparison.
- **Single dataset:** no independent simulation, time interval, or particle-sample validation was performed.
- **Representation choices:** magnitude spectrograms, resizing, selected channels, and sampling affect the discovered groups.
- **Scale-up:** million-particle runtimes are extrapolations under a linear assumption.

Careful claim

The results support a useful way to organize these records, not a claim that the five groups are validated physical categories.

Conclusions and next steps

1. A six-channel spectrogram pipeline was implemented for 12,800 PIC particle trajectories.
2. The spectrogram CNN + DEC model learned 32D features and produced five clusters.
3. It achieved the highest observed silhouette (0.87985) among the three tested methods, under internal evaluation.
4. The workflow offers an indexing and exploration tool for subsequent domain-specific analysis.

Next steps

Validate across additional simulations; relate clusters to physical diagnostics; evaluate sensitivity to channels and preprocessing; improve throughput for large datasets.

Thank you

Questions and discussion

Linshuo CHEN | Supervisor: Andrea CIARDI